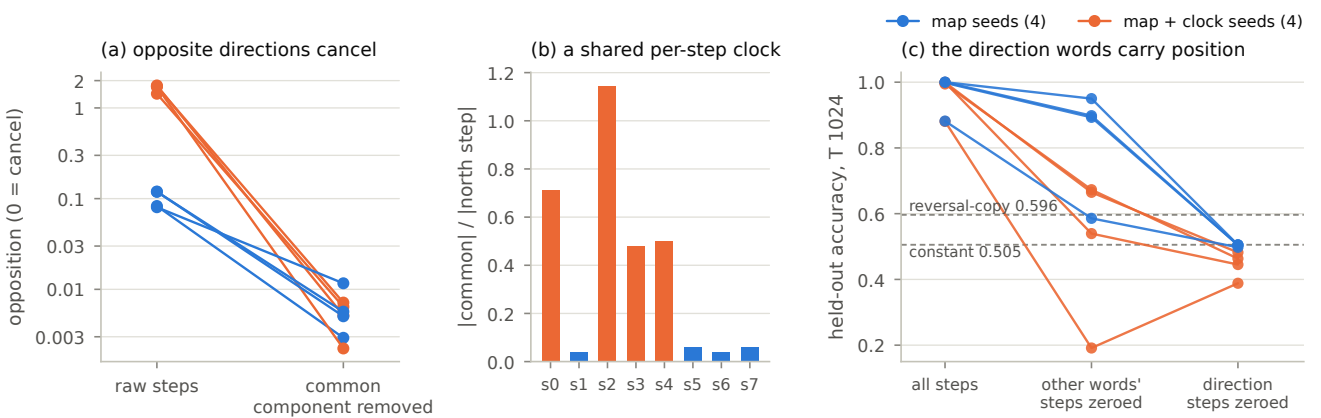
map seeds (4)
map + clock seeds (4)
(a) opposite directions cancel
opposition (0 = cancel)
2
1
0.3
0.1
0.03
0.01
0.003
raw steps
common component removed
(b) a shared per-step clock
|common| / |north step|
1.2
1.0
0.8
0.6
0.4
0.2
0.0
s0 s1 s2 s3 s4 s5 s6 s7
(c) the direction words carry position
held-out accuracy, T 1024
1.0
0.8
0.6
0.4
0.2
reversal-copy 0.596
constant 0.505
all steps
other words' steps zeroed
direction steps zeroed